MRA Project Part B

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Agenda – Part B: Market Basket Analysis Using Association Rules

- **Business Context**: Overview of the grocery retail environment and need for basket-level insights
- **Problem Statement & Objectives** Define the analytical challenge and goals—identifying item combinations to enhance sales
- **Data Understanding** Structure and features of POS transactional data
- **Methodology** Approach for preprocessing, transaction grouping, and rule mining
- **Rule Mining Results** Presentation of top frequent item sets and association rules with support, confidence, and lift
- **Insights from Item Combinations** Interpretation of results—product affinities, complementary items, and strategic pairing
- **Marketing Strategy Recommendations** Suggested combo offers, discount strategies, and placement tactics based on insights
- **KNIME Workflow Snapshot** Overview of the workflow used for mining and visualization
- **Conclusion & Business Impact** Summary of findings and how they support profitability, loyalty, and operational efficiency

Business Overview

In the highly competitive grocery retail industry, understanding customer buying patterns is crucial for enhancing sales, increasing customer satisfaction, and improving profitability. By identifying frequently purchased item combinations, grocery stores can craft effective marketing strategies, optimize inventory management, and tailor promotions to meet customer needs. Leveraging Point of Sale (POS) data can unlock valuable insights that drive customer-centric offerings, such as combo packs, discounts, and targeted promotions, which can increase basket size and improve customer retention. This analysis aligns with business goals by maximizing revenue, reducing operational costs, and boosting customer loyalty.

Problem Statement

Objective:

As a business analyst, the goal is to analyze the POS transactional data to identify frequently purchased item combinations. Using association rule mining or similar techniques, the aim is to uncover patterns that will help the store create targeted combo offers and discounts, ultimately driving revenue growth by increasing customer purchases and average basket size.

Data Description

The dataset provided contains three years of transactional data from the company, with each row representing a unique order. Below is an explanation of the key attributes:

Date: The date when the transaction took place.
Order_id: A unique identifier for each customer order.
Product: The individual item purchased in the transaction.

EDA STARTS

	Date	Order_id	Product
0	01-01-2018	1	yogurt
1	01-01-2018	1	pork
2	01-01-2018	1	sandwich bags
3	01-01-2018	1	lunch meat
4	01-01-2018	1	all- purpose
5	01-01-2018	1	flour
6	01-01-2018	1	soda
7	01-01-2018	1	butter
8	01-01-2018	1	beef
9	01-01-2018	1	aluminum foil

	Date	Order_id	Product
20631	25-02-2020	1138	all- purpose
20632	25-02-2020	1138	sandwich bags
20633	25-02-2020	1138	toilet paper
20634	25-02-2020	1138	soda
20635	25-02-2020	1138	soda
20636	25-02-2020	1138	soda
20637	25-02-2020	1138	paper towels
20638	26-02-2020	1139	soda
20639	26-02-2020	1139	laundry detergent
20640	26-02-2020	1139	shampoo

This is Head and Tail of the provided Sales Data

EDA cont..

```
(20641, 3)
```

The dataset has 20641 rows and 3 columns.

```
Date        0
Order_id    0
Product     0
dtype: int64
```

- There are no null values in the dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20641 entries, 0 to 20640
Data columns (total 3 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Date      20641 non-null  object
 1   Order_id  20641 non-null  int64
 2   Product   20641 non-null  object
dtypes: int64(1), object(2)
memory usage: 483.9+ KB
```

- All the Datatypes are correct no intervention is required.
- Duplicates in the data, which need to be grouped
- No Missing values in the data

EDA cont..

- **Dataset Size:**
  - A total of **20,641 product-level transactions** across **603 unique dates** provides a substantial and well-spread foundation for basket analysis.
- **Order-Level Insights**
  - There are approximately **1,139 unique orders**, with a fairly even distribution:
    - **Median Order ID: 581**,
    - **25% of orders below 292**, and
    - **Top quartile above 862**.
  - This range reflects a healthy mix of new and returning customers over time.
- **Temporal Demand Pattern**
  - The most active day is **08-02-2019**, with **183 items sold**—suggesting a spike worth further investigation (campaign, promotion, seasonality?).
- **Product Popularity**
  - **37 unique products** are sold, with **poultry** being the most frequent, appearing **640 times**—making it a strong candidate for combo deals or feature placement in promotions.

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Date	20641	603	08-02-2019	183	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Order_id	20641.000	NaN	NaN	NaN	575.986	328.557	1.000	292.000	581.000	862.000	1139.000
Product	20641	37	poultry	640	NaN	NaN	NaN	NaN	NaN	NaN	NaN

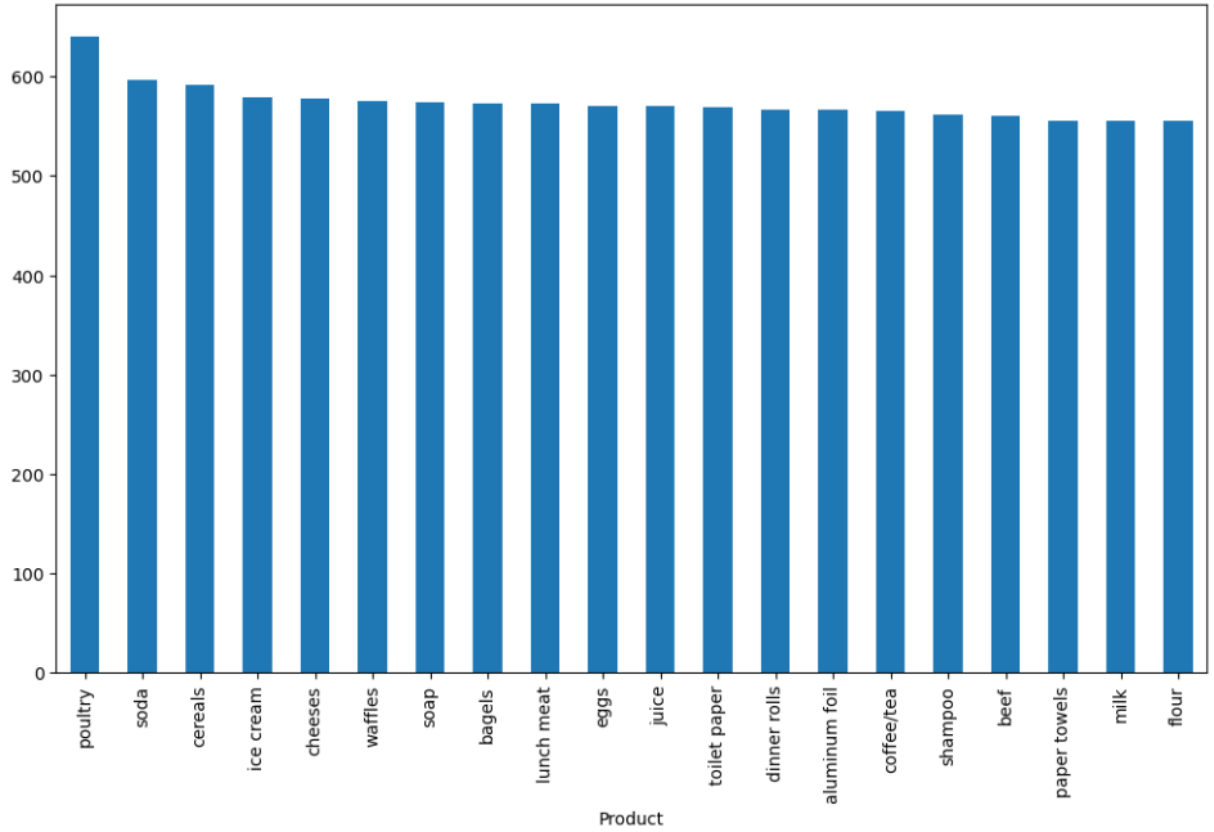
EDA cont..

•**Top-Selling Item**: *Poultry* leads significantly with over **640 units sold**, establishing it as the most frequently purchased product—an excellent candidate for promotional combos or feature placement.

•**High-Demand Staples**: Products like *soda, cereals, ice cream, cheeses,* and *waffles* also show strong sales volumes (**500–600 units**), suggesting they form the core of recurring shopping baskets.

•**Diverse Basket Composition**: A wide range of items—spanning **breakfast (bagels, eggs), beverages (juice, coffee/tea), and hygiene (soap, toilet paper, shampoo)**—indicates broad category engagement and cross-selling potential.

•**Marketing Opportunity**: Frequent co-occurrence of staple goods and household items opens the door for **multi-category combo deals** (e.g., pairing beverages with breakfast items or hygiene products with pantry staples).



EDA cont..

- **Frequent Co-occurrences Across Orders**
  - Core staples like *bagels, beef, butter, shampoo, soap, soda,* and *toilet paper* appear consistently across multiple orders, indicating they are **commonly bundled**.

- **Strong Pairing Candidates**
  - Items such as *dishwashing liquid*, *spaghetti sauce*, *sugar*, and *shampoo* are often purchased together in orders with *dinner rolls* or *all-purpose flour*. These suggest **natural household or pantry groupings** for combo offers.

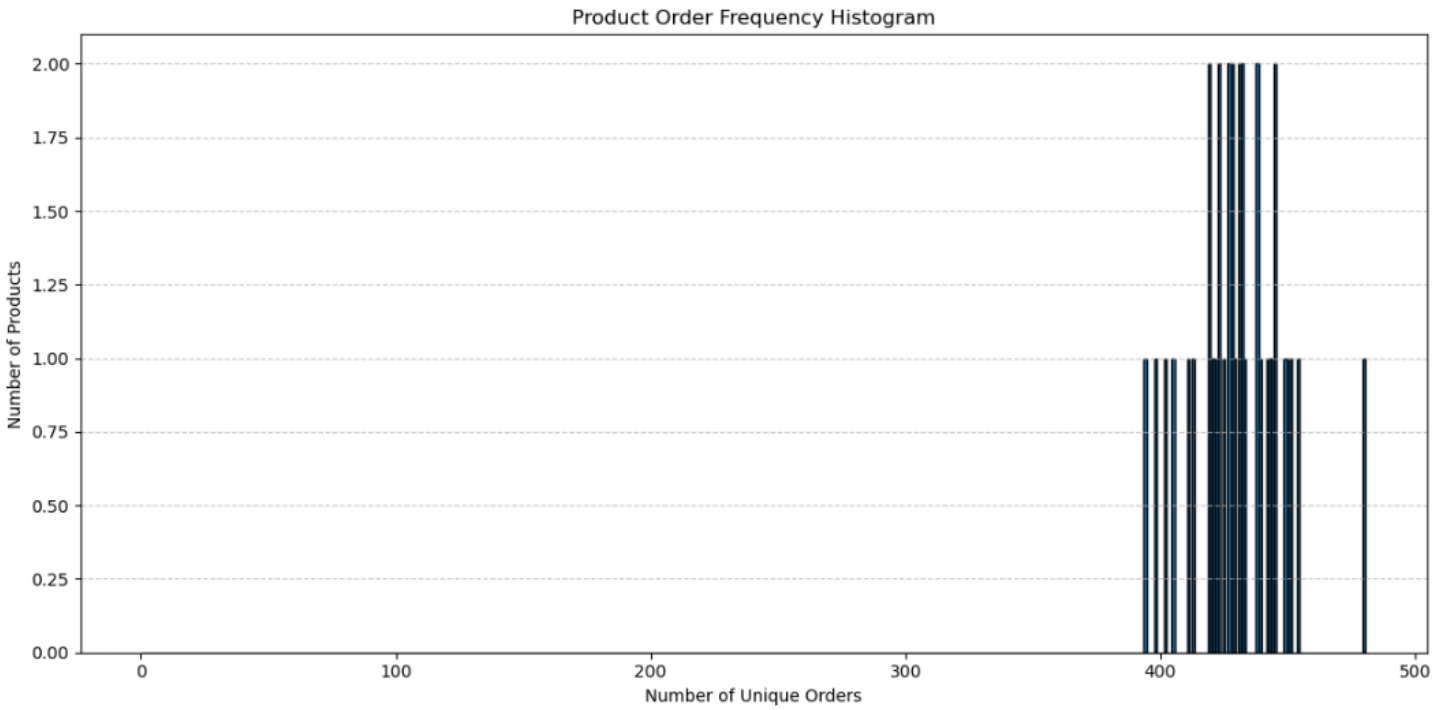
- **Core Convenience Items**
  - Products like *aluminum foil* and *coffee/tea* tend to appear with a wide variety of other items, implying their **versatility across baskets**—ideal for promotional anchoring.

- **Cross-Category Basket Behavior**
  - Presence of hygiene items (*shampoo, soap*) with grocery goods points to **multi-category purchasing behavior**, making them prime candidates for cross-category bundling (e.g., “meal & hygiene” kits).

Product	all-purpose	aluminum foil	bagels	beef	butter	cereals	cheeses	coffee/tea	dinner rolls	dishwashing liquid/detergent	...	sandwich loaves	shampoo	soap	soda	spaghetti sauce	sugar	toi pap
Order_id																		
1	3.000	1.000	0.000	1.000	1.000	0.000	0.000	0.000	2.000	0.000	...	0.000	1.000	1.000	1.000	0.000	0.000	0.0
2	0.000	1.000	0.000	0.000	0.000	1.000	1.000	0.000	0.000	1.000	...	0.000	2.000	0.000	0.000	0.000	0.000	1.0
3	0.000	0.000	1.000	0.000	0.000	1.000	1.000	0.000	1.000	0.000	...	1.000	1.000	2.000	1.000	3.000	0.000	2.0
4	1.000	0.000	0.000	0.000	0.000	1.000	0.000	0.000	0.000	0.000	...	0.000	0.000	0.000	1.000	0.000	0.000	1.0
5	1.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	1.000	0.000	...	1.000	0.000	0.000	1.000	1.000	0.000	1.0

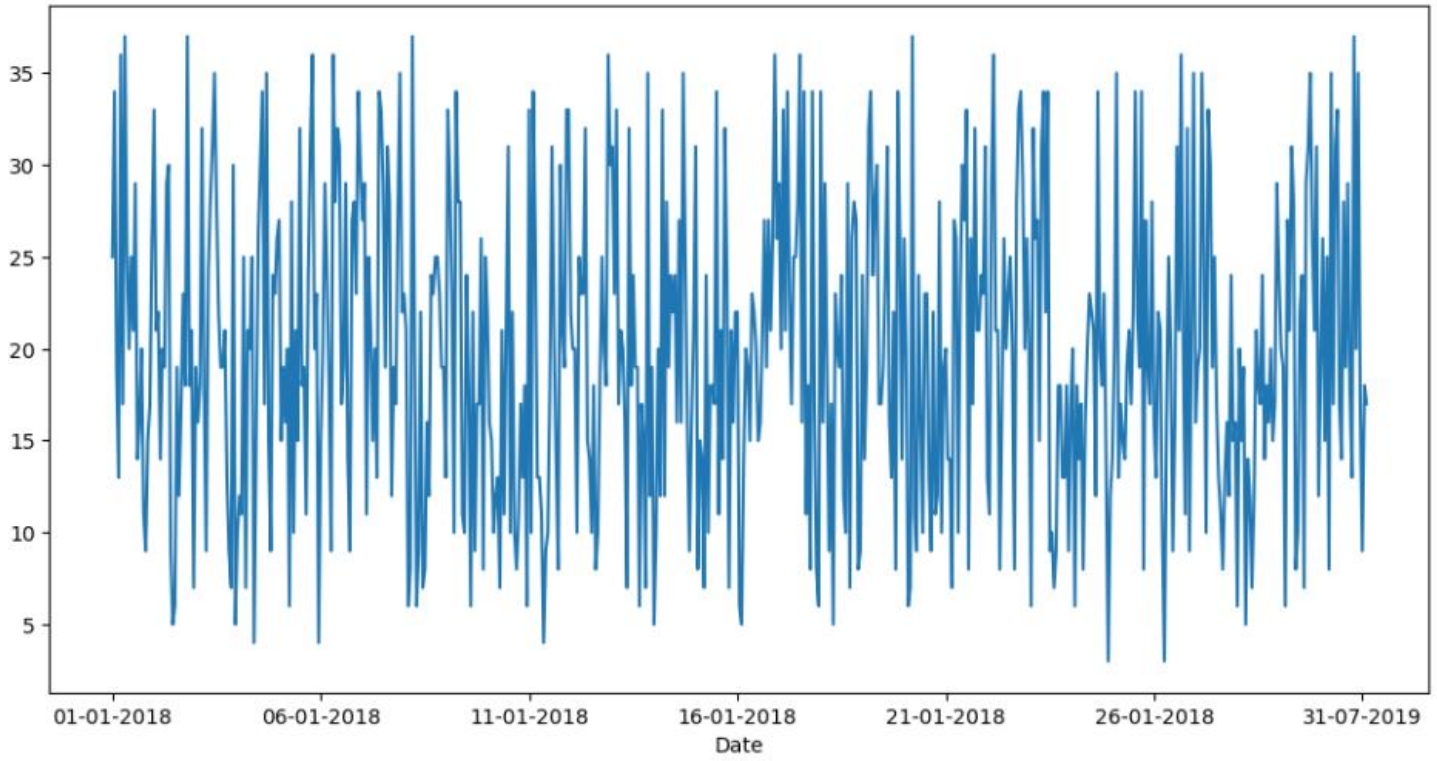
EDA cont..

- **High Frequency Core Products**
  - Most products appear in **380 to 450 unique orders**, suggesting a strong backbone of consistently purchased items.
- **Balanced Distribution**
  - The histogram shows a relatively tight spread, meaning most products have **comparable exposure across the order base**



EDA cont..

- **Stable Variety with Seasonal Peaks**
  - Most days show a steady flow of **10 to 25 unique products**, indicating consistent diversity in daily shopping baskets.
  - **Spikes nearing 30+ products** suggest promotional campaigns, holidays, or restocking events driving broader basket composition.
- **Operational Consistency**
  - There's no prolonged dip or outage in variety, which indicates strong **supply chain consistency** and **steady customer engagement** across the timeframe.
- **Trend Opportunity**
  - Sharp one-day peaks could be mined to identify **what drove the variety**?



EDA SUMMARY [INFERENCES]

- **Dataset Overview:**
  - **Transactions**: 20,641 rows across 603 unique dates
  - **Orders**: 1,139 unique Order IDs
  - **Product Assortment**: 37 distinct products spanning groceries, beverages, hygiene, and household items

- **Top-Selling Products**
  - Poultry, Soda, Ice Cream, Toilet Paper, and Shampoo dominate unit sales—each surpassing 500 units
  - These are the core basket drivers and ideal anchor items for combo promotions

- **Product Penetration**
  - Most products appear in 380–450 unique orders, indicating balanced, high-penetration inventory.
  - Histogram shows tight distribution with minimal deadweight—inventory strategy appears optimized

- **Daily product variety** ranges from 10–30 items, showing consistent customer engagement
  - Peak sales day (08-Feb-2019) saw 183 units sold—an event/campaign worth replicating

- **Item-Level Transaction Patterns**
  - Heatmap reveals frequent co-occurrence of pantry and hygiene items (e.g., Butter + Soap, Toilet Paper + Shampoo)
  - Clear signs of multi-category basket building behavior

MBA Starts

Process Overview: Association Rules in Context

1. Data Preparation

- Cleaned transaction-level POS data containing Order ID and Product Name
- Transformed into basket format where each row represents one transaction's itemset

2. Rule Mining Technique

- Metrics used for evaluating rules: **Support, Confidence, and Lift.**

3. Association Rule Mining Parameters (KNIME)

Minimum Support: **0.05** → Only includes item combinations that appear in at least 5% of all transactions (i.e., sufficiently frequent to be business-relevant)

Minimum Confidence: **0.4** → Rules are retained only if there's at least a 40% chance that the consequent is purchased when the antecedent is present.

- **Maximal Itemset Length**: **10** → Allows mining of both short and long rules to capture complex combinations
- **Underlying Structure**: **ARRAY**, optimized for basket-format datasets
- **Transaction Column**: Product_SplitResultSet, identifying items grouped per order

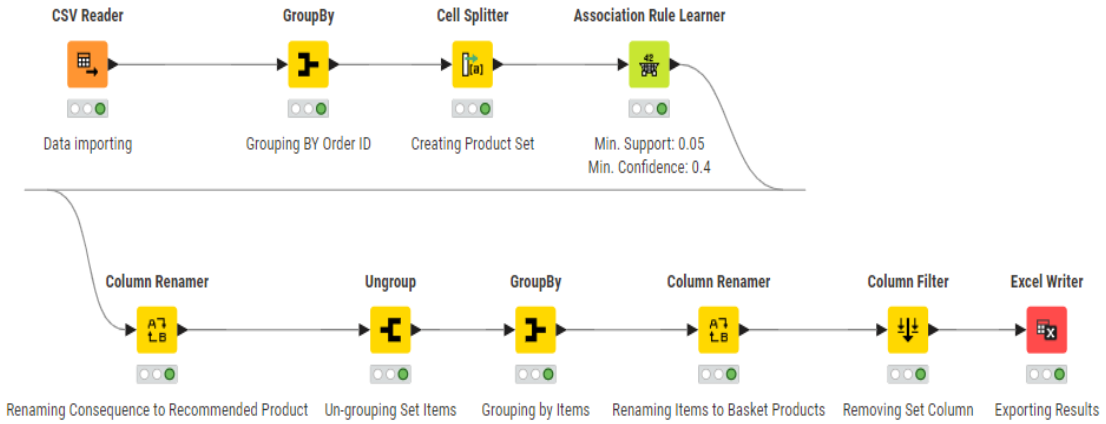
MBA (using KNIME)

KNIME Workflow:

Output table

Rows: 755 | Columns: 6

#	RowID	Support Number (double)	Confidence Number (double)	Lift Number (double)	Rec_Product String	implies String	Basket_Product String
1	Row0	0.07	0.448	1.177	cheeses	←	all- purpose, aluminum foil, all- purpose, aluminum
2	Row1	0.066	0.437	1.155	dinner rolls	←	all- purpose, bagels, all- purpose, bagels, all- purpo
3	Row2	0.064	0.443	1.169	sandwich loaves	←	all- purpose, beef, all- purpose, beef, all- purpose, t
4	Row3	0.067	0.446	1.177	pork	←	all- purpose, cereals, all- purpose, cereals, all- purp
5	Row4	0.066	0.449	1.183	laundry detergent	←	all- purpose, coffee/tea, all- purpose, coffee/tea, a
6	Row5	0.063	0.442	1.171	beef	←	all- purpose, flour, all- purpose, flour, all- purpose, f
7	Row6	0.061	0.446	1.177	sandwich bags	←	all- purpose, hand soap, all- purpose, hand soap, a
8	Row7	0.062	0.447	1.18	flour	←	all- purpose, individual meals, all- purpose, individ
9	Row8	0.069	0.45	1.176	cheeses	←	all- purpose, juice, all- purpose, juice, all- purpose,
10	Row9	0.069	0.448	1.182	shampoo	←	all- purpose, ketchup, all- purpose, ketchup, all- pu
11	Row10	0.072	0.444	1.168	flour	←	all- purpose, laundry detergent, all- purpose, laund
12	Row11	0.071	0.446	1.179	mixes	←	all- purpose, lunch meat, all- purpose, lunch meat,
13	Row12	0.068	0.447	1.174	coffee/tea	←	all- purpose, milk, all- purpose, milk, all- purpose, n
14	Row13	0.068	0.449	1.182	butter	←	all- purpose, mixes, all- purpose, mixes, all- purpos



MBA (using KNIME) Inferences

In-Depth Insights

1: Core Basket Drivers Involve Common Household Essentials: From your dataset, many of the frequently occurring items in the baskets include:

- all-purpose cleaner
- aluminum foil
- coffee/tea
- Cereals
- bagels
- beef

These products are essential or staple items that are **frequently bought** and likely form the **base of shopping baskets**. This suggests:

- Customers tend to stock up on **both food and non-food essentials** in the same trip.
- These staple items act as **"basket anchors"**, influencing purchase of secondary items.

Implication: These items can be used as “hooks” to trigger upselling via paired promotions.

2: Cheeses, Pork, and Laundry Detergent Are High-Influence Association Targets:

The following items were frequently purchased alongside large baskets of mixed goods, with:

- **High confidence (~45%)**: Customers who purchase certain baskets have a ~45% chance of also purchasing these.
- **Lift > 1.17**: Strong positive association beyond random chance.

Recommended Product	Average Confidence	Average Lift	Implication
Cheeses	44.8%	1.18	Strong complementary item
Pork	44.6%	1.17	High protein + grocery synergy
Laundry Detergent	44.9%	1.18	Strong correlation between cleaning and grocery

MBA (using KNIME) Inferences

3: Certain Product Combinations Suggest Lifestyle or Meal Preferences: Some baskets hint at **specific customer behaviors**, for example:

- **All-purpose + Coffee/Tea → Laundry Detergent**
  Suggests household restocking missions, not just food-based trips.
- **Beef + All-purpose → Sandwich Loaves**
  Indicates prep for meals like sandwiches, implying convenience-oriented buying.
- **Bagels + All-purpose → Dinner Rolls**
  Suggests customers interested in **breakfast and bread-focused meals**.

Implication: You can derive **purchase intent** (e.g., meal planning, household cleaning) from these rules and tailor **time-based offers** (e.g., weekend family packs).

Recommendations:

1. Launch Cross-Category Bundle Offers: Design cross-category promotions such as:

- **"Weekend Essentials Combo"**: All-purpose + Coffee/Tea + Laundry Detergent (Home care + Grocery)
- **"Protein + Bread Value Pack"**: Beef + Sandwich Loaves
- **"Morning Meal Bundle"**: Bagels + Dinner Rolls + Cheese

2. Optimize Store Layout and Product Placement

- Use physical proximity in the store and UI proximity on e-commerce platforms:
- Co-locate **all-purpose cleaner and cheeses** with signage like “Perfect Pairings for the Week.”
- Place **coffee/tea, aluminum foil**, and **detergents** in a restock-oriented aisle zone.
- Put **bagels, sandwich loaves**, and **beef** close together to encourage meal-focused shopping.